INCH-POUND

MIL-DTL-7077D
16 May 2011
SUPERSEDING
MIL-DTL-7077C
20 September 2007

DETAIL SPECIFICATION

INDICATOR, TEMPERATURE, ELECTRICAL RESISTANCE, MULTIFUNCTION, -70 ºC TO 150 ºC, 2-INCH SIZE

Inactive for new design after 24 March 1999.

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

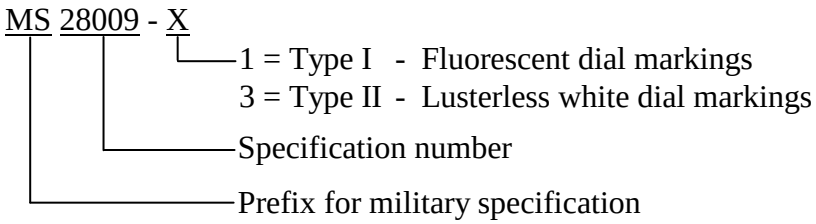
1.1 Scope. This specification covers multifunction electrical resistance temperature indicators.

1.2 Classification.

1.2.1 Type. The electrical resistance temperature indicators are of the following types, as specified below and in the associated specification sheets (see 6.2).

Type I - Fluorescent dial markings (MS28009-1)
Type II - Lusterless white dial markings (MS28009-3)

1.3 Part or identifying number (PIN). The PIN to be used for landing gear position indicators acquired to this specification is shown in MS28009.



Comments, suggestions, or questions on this document should be addressed to Defense Logistics Agency - Aviation, ATTN: DLA Aviation-VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616 or e-mailed to STDZNMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at https://assist.daps.dla.mil.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

FEDERAL STANDARDS

FED-STD-595/37875	- Miscellaneous, Flat or Lusterless
FED-STD-595/37038	- Miscellaneous, Flat or Lusterless

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-I-7070	- Indicator, Temperature, Electrical Resistance, Hermetically Sealed, General Specification for.
MS28009	- Indicator - Temperature, Electrical Resistance Multi-function, -70 to +150 C, 2-Inch Size.
MS33558	- Numerals and Letters, Aircraft Instrument Dial, Standard Form of.
MS33639	- Cases, Instrument, Clamp-Mounted, Aircraft.
MS33678	- Connector, Receptacle, Electrical, Integral Mounting.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	- Identification Marking of U.S. Military Property.

(Copies of these documents are available at https://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

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2.2.2 Other government documents, drawings, and publications. The following other government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

STANDARDIZATION DOCUMENTS

SD-6 - Provisions Governing Qualification.

(Copies of this document are available at https://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheet. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 Qualification. Indicators furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.4 and 6.3).

3.3 General specification. The requirements of MIL-I-7070 apply as requirements of this specification with the exceptions and additions specified herein. In the event the two specifications conflict, the requirements of this specification shall govern.

3.4 Construction.

3.4.1 Case. The case shall be in accordance with MS33639 as shown on MS28009.

3.4.2 Scale length. The scale of the indicator shall be as long as practicable and shall not be less than 1-3/4 inches as measured on the outer arc of the shortest graduation.

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3.4.3 Pointer. The pointer design shall be as shown on figure 1.

3.4.3.1 Pointer overswing. The pointer overswing shall not be less than 3 percent of full scale angular deflection beyond each end of the scale.

3.4.4 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements and promotes economically advantageous life cycle costs.

3.5 Marking. The markings shall be durable and in accordance with figure 1. The flat or lusterless white shall be color 37875 of FED-STD-595. The flat or lusterless black shall be color 37038 of FED-STD-595. The form of the letters shall conform to MS33558.

3.6 Electrical receptacle. An electrical receptacle, MS33678-14S-2P, shall be provided and located as shown on MS28009. Receptacle terminals shall be identified as follows:

Terminal	Connection
A	+24 volt battery
B	Resistance bulb
C	Common-negative 24-volt battery and remaining resistance bulb connection
D	Open

3.7 Weight. The weight of the complete indicator shall not exceed 1.4 pounds.

3.8 Identification.

3.8.1 Identification of products. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.8.2 Instrument identification decals. The indicator shall be provided with one each of the two identification decals as shown on figure 2. The letters shall be finished lusterless white or fluorescent to correspond to the dial finish. The background of the decal shall be transparent or lusterless black. The letters shall be 0.125 inch ± 0.010 inch high and the width of the lines shall be 0.20 ± 0.005 inch. The form of the letters shall conform to MS33558. The decal shall be the water-applied type and shall be suitable for application to the cover glass of the indicator. When mounted on the cover glass of the indicator, the decal shall be capable of meeting the tests specified in section 4 without peeling, blistering, fading, loss of adhesion, cracking, softening, or hardening. Four hours after application, the decals shall adhere sufficiently to the cover so that the surface shall withstand normal handling. Each set of decals shall be packaged in an envelope marked "THIS ENVELOPE CONTAINS IDENTIFICATION DECAL." Complete and detail instructions for mounting the decal on the cover glass of the indicator shall be furnished with each set of decal.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.4).
b. Conformance inspection (see 4.5).

4.2 Test conditions.

4.2.1 Test voltage. Unless otherwise specified herein, the test voltage shall be 28.5 ± 0.5 volts (V) direct current (DC), unless otherwise specified.

4.3 Test tolerances. The tolerances specified in table I that are not enclosed in parenthesis are applicable only when the indicators are subjected to all the tests in the sequence specified herein. The tolerances specified in table I that are enclosed in parenthesis are applicable only when the indicators are subjected to a specific test independent of sequence.

4.4 Qualification inspection. The qualification inspection shall consist of all the tests specified in 4.6 and in MIL-I-7070.

4.5 Conformance inspection. The conformance inspection shall consist of the individual tests of 4.5.1, sampling plan A inspections of 4.5.2.1, and sampling plan B inspections of 4.5.2.2, as applicable (see 6.2).

4.5.1 Individual tests. Each indicator shall be subjected to the following tests.

a. Scale error at room temperature (see 4.6.1).
b. Position error (see 4.6.2).
c. Friction error (see 4.6.3).
d. Insulation leakage (see 4.6.4).

4.5.2 Sampling plan inspections.

4.5.2.1 Sampling plan A. One indicator shall be selected at random from each 100 or less produced on the acquisition order and subjected to the following tests.

a. Individual tests (see 4.5.1).
b. Reduced voltage operation (see 4.6.5).
c. Magnetic effect (see 4.6.6).
d. Resistance bulb current (see 4.6.7).
e. Vibration error (see 4.6.8).

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4.5.2.2 Sampling plan B. Unless otherwise specified (see 6.2), three indicators shall be selected at random from the first 15 produced on the acquisition order and subjected to the following tests.

a. Sampling plan A tests (see 4.5.2.1).
b. Response time (see 4.6.9).
c. Damping factor (see 4.6.10).
d. Power consumption (see 4.6.11).
e. Bulb lead resistance (see 4.6.12).

4.6 Test methods.

4.6.1 Scale error at room temperature. The test resistance specified in table I shall be applied to the resistance bulb terminals of the indicator. The test voltage shall be applied and the scale errors at the test points specified in table I shall not exceed the tolerances specified in table I.

4.6.2 Position error. When the indicator is subjected to the position error test, the maximum change in indication shall not exceed 4 ºC.

4.6.3 Friction error. When the indicator is subjected to the friction test, the maximum difference in reading before and after tapping shall not exceed 4 ºC.

4.6.4 Insulation leakage. When the indicator is subjected to the insulation leakage test, the insulation resistance shall be a minimum of 20 megohms.

4.6.5 Reduced voltage operation. The indicator shall be subjected to a scale error at room temperature test with an applied voltage of 22.5 V at the test points specified in table I. The indications at the reduced voltage shall not differ from the indications with 28.5 V applied by more than the tolerances specified in table I.

4.6.6 Magnetic effect. When the indicator is subjected to the magnetic effect test, it shall not cause the magnetic compass to deflect more than 5 degrees.

4.6.7 Resistance bulb current. A total resistance of 100 ohms shall be applied across the bulb circuit terminals of the indicator, the test voltage applied, and the bulb circuit current measured. The current consumed in the resistance bulb circuit shall not exceed 22 milliampere.

4.6.8 Vibration error. When the indicator is subjected to the vibration error test, the maximum amplitude of pointer oscillation throughout the frequency range shall not exceed 4 ºC. The maximum difference between the original position of the pointer during vibration shall not exceed 4 ºC. Pointer oscillation shall not be considered as pointer variation.

4.6.9 Response time. When the indicator is subjected to the response time test, the response time shall not exceed 3 seconds.

4.6.10 Damping factor. When the indicator is subjected to the damping factor test, the damping factor shall not be less than 1.6.

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4.6.11 Power consumption. When the indicator is subjected to the power consumption test, the power consumption shall not exceed 1.5 watts.

4.6.12 Bulb lead resistance. The indicator shall be connected as shown on figure 3 and shall be subjected to a scale error at room temperature test. The indicator with bulb lead resistance shall not differ from the indication without bulb resistance by more than the tolerances specified in table I.

4.7 Maintenance of qualification. At specified intervals determined by the qualifying activity, the manufacturer must be able to demonstrate that the company still has the capabilities and facilities necessary to produce the QPL items in accordance with this specification and the provisions governing qualification specified in SD-6.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the inventory control point's packaging activities within the military service or defense agency, or within the military service's system command. Packaging data retrieval is available from the managing military department's or defense agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The electrical resistance temperature indicators covered by this specification are intended for use in aircraft to indicate cabin or free air temperature.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type and part number of transmitter desired (see 1.2).
c. The specific issue of individual documents referenced (see 2.2.1 and 2.2.2).
d. Sampling plan A or B, as required (see 4.5).
e. Alternate quantity for sampling plan B inspections, if required (see 4.5.2.2).
f. Packaging (see 5.1).

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6.3 Qualification. With respect to products requiring qualification, awards will be made only for products that are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-7077 whether or not such products have actually been listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the federal government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Logistics Agency Aviation, Attn: DLA Aviation-VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297 or STDZNMGT@dla.mil.

6.4 Bezel. The temperature indicator covered by this specification can be used on instrument panels requiring MS33638 case mounting by the application of an instrument mounting bezel in accordance with MS28055.

6.5 Subject term (key word) listing.

air
aircraft
cabin

6.6 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

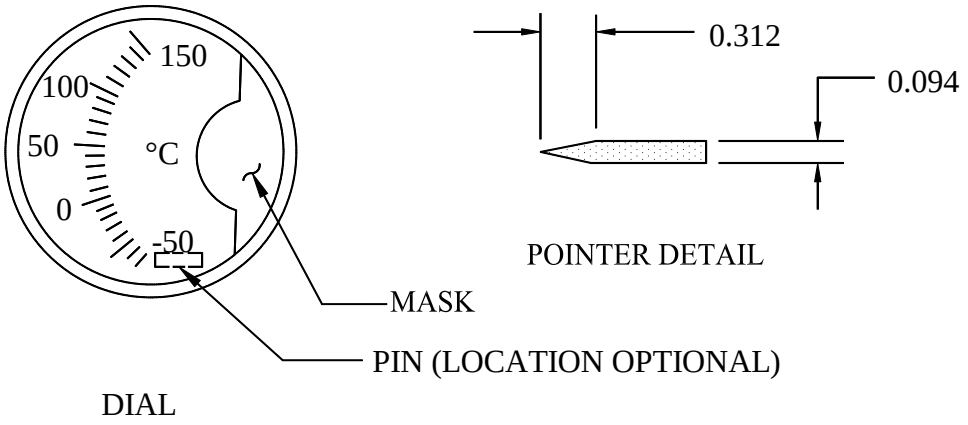
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TABLE I. Test tolerance.

Test point (ºC)	Test resistance (Ohms)	Scale error at room temperature (tolerance ± ºC)	Reduced voltage (tolerance ± ºC)	Scale error at low temperature (tolerance ± ºC)	Scale error at high temperature (tolerance ± ºC)	Scale error at room temperature after open and short circuit (tolerance ± ºC)	Scale error at room temperature after vibration failure (tolerance ± ºC)
-70	68.27	3.0 (3.0)	2.0 (2.0)	7.0 (7.0)	7.0 (7.0)	10.0 (4.0)	10.0 (7.0)
-30	80.56	3.0 (3.0)	2.0 (2.0)	6.0 (6.0)	6.0 (6.0)	10.0 (4.0)	10.0 (7.0)
0	90.38	2.0 (2.0)	2.0 (2.0)	4.0 (4.0)	4.0 (4.0)	10.0 (3.0)	10.0 (6.0)
30	100.91	3.0 (3.0)	2.0 (2.0)	6.0 (6.0)	6.0 (6.0)	10.0 (4.0)	10.0 (7.0)
80	120.36	3.0 (3.0)	2.0 (2.0)	6.0 (6.0)	6.0 (6.0)	10.0 (4.0)	10.0 (7.0)
120	137.78	3.0 (3.0)	3.0 (3.0)	6.0 (6.0)	6.0 (6.0)	10.0 (4.0)	10.0 (7.0)
150	151.91	3.0 (3.0)	3.0 (3.0)	7.0 (7.0)	7.0 (7.0)	10.0 (4.0)	10.0 (7.0)

Test point (ºC)	Test resistance (Ohms)	Extreme low temperature exposure		Scale error at room temperature after high temperature exposure (tolerance ± ºC)	Scale error with bulb lead resistance (tolerance ± ºC)
		Scale error at low temperature (tolerance ± ºC)	Scale error at room temperature after low temperature exposure (tolerance ± ºC)		
-70	68.27	14.0 (7.0)	10.0 (3.0)	10.0 (3.0)	3.0 (3.0)
-30	80.56	13.0 (6.0)	10.0 (3.0)	10.0 (3.0)	3.0 (3.0)
0	90.38	12.0 (4.0)	10.0 (2.0)	10.0 (2.0)	2.0 (2.0)
30	100.91	13.0 (6.0)	10.0 (3.0)	10.0 (3.0)	3.0 (3.0)
80	120.36	13.0 (6.0)	10.0 (3.0)	10.0 (3.0)	3.0 (3.0)
120	137.78	13.0 (6.0)	10.0 (3.0)	10.0 (3.0)	3.0 (3.0)
150	151.91	14.0 (7.0)	10.0 (3.0)	10.0 (3.0)	3.0 (3.0)

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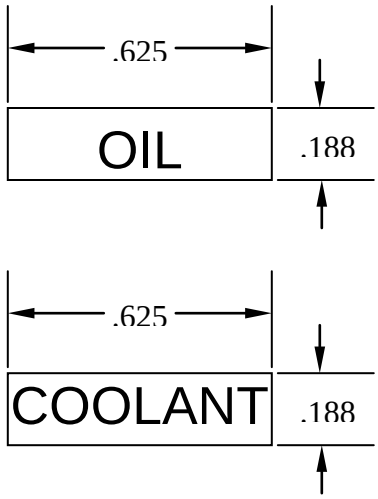
DIAL

Marking	Height or length ±0.005	Width of line or graduation ±0.005	Material or finish
All numerals	0.125	0.020	Fluorescent luminescent material or flat or lusterless white
Major graduations	0.156	0.031	
Minor graduations	0.093	0.020	
Lettering "ºC"	0.125	0.020	
Shaded portion of pointer	--	--	
PIN	0.062	0.010	
Background of dial	--	--	Flat or lusterless black

Dimensions in inches.
Unless otherwise specified, tolerances: ±.031

FIGURE 1. Dial, pointer, and markings.

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Dimensions in inches.
Tolerances: ±.031.

FIGURE 2. Decal size and marking.

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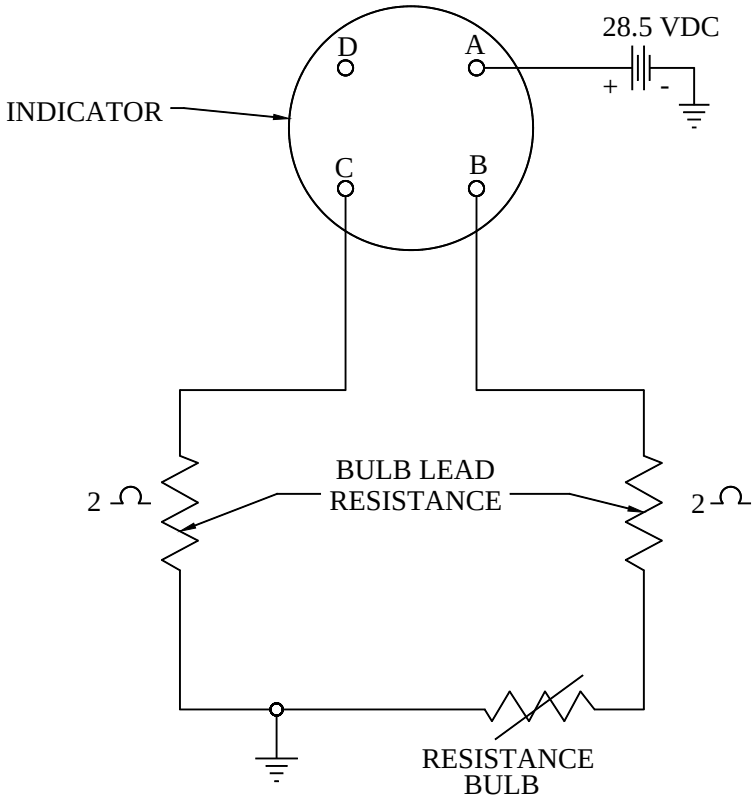


FIGURE 3. Wiring diagram.

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Air Force - 99

Preparing Activity:
DLA - GS1

(Project 6685-2011-009)

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